

SPIDEYOS

Android Spider-Man Experience

A companion suite for Android 15+ phones & tablets

Launcher · Gemini Assistant · Web Notifications · Messages · Mail · Peter
Camera

Case Study by **Jaikar**

Open Source · No Root · Snapdragon & MediaTek

Unofficial fan / inspired project. Not affiliated with Marvel Entertainment or Sony.

1. The Idea

What if your Android phone didn't just *look* like Spidey — what if Spidey *lived* on it? SpideyOS turns the device into a character-driven companion: the home screen wears a web motif, notifications arrive in Spidey's voice, and a Gemini-powered assistant greets you by name.

“Hey Jaikar, you’ve got mail.”

The product goal is presence: a personal assistant that swings through messages, mail, camera, and system alerts — not a static theme pack.

2. The Problem

Stock Android is powerful but emotionally flat. Icon packs change colors; they don't change how the phone talks to you. Generic assistants sit in a separate app and ignore the rest of the system. Fans want immersion; engineers want installable software that works without rooting every OEM skin.

- Themes don't speak — they only recolor.
- Assistants are generic and siloed from notifications, mail, and camera.
- Custom ROMs exclude most users and break OEM updates.
- Portfolio projects rarely show end-to-end system UX + AI product thinking.

3. The Solution

SpideyOS is a single sideloadable APK companion suite for Android 15 (API 35)+. No custom ROM. No Magisk required. Users set Spidey Launcher as home, grant notification and overlay permissions, and Spidey becomes part of daily flow.

Module	What it does
Spidey Launcher	Web-themed home, app grid, open animations
Spidey Assistant	Floating overlay + Gemini persona for Jaikar
Notification Rewriter	Spidey-voiced alerts for mail & messages
Web Messages	Spider-web bubbles + send-line animation
Mail Digest	New mail announced by Spidey
Peter Camera	CameraX comic UI + Spidey photo reactions

4. Architecture

SpideyOS is modular inside one application process. The launcher hosts navigation; the assistant overlay and notification listener run as services; feature modules (messages, mail, camera) share theme tokens and a user profile (default display name: **Jaikar**).

Data flow (simplified)

User → Launcher → Modules

Spidey Overlay → Gemini API (persona system prompt)

System Notifications → NotificationListener → Spidey voice copy → UI / overlay

Mail / Messages → Digest → Spidey announcements

CameraX capture → Spidey reaction prompts

This architecture prioritizes universal installability over full OS replacement — the honest path to “works on all Android 15+ phones and tablets.”

5. Feature Walkthrough

Spidey voice notifications

With notification access enabled, SpideyOS rewrites selected alerts into character: “Spidey here — Jaikar, you’ve got mail.” Fun first, then actionable — tap through to the source app or Mail Digest.

Web messages

Message bubbles use web-inspired geometry and a web-line send animation so conversation feels kinetic — matching the launcher’s motion language.

Peter Camera

A CameraX viewfinder with comic-frame overlays and a web shutter. After capture, Spidey can react (“Nice shot — Daily Bugle material?”) via the assistant pipeline.

6. Tech Stack & Android 15

Minimum SDK: **Android 15 (API 35)**. Language: **Kotlin**. UI: **Jetpack Compose** + Material 3. AI: **Google Gemini API**. Camera: **CameraX**. Persistence: **DataStore**. CI: **GitHub Actions** for debug/release APK artifacts.

Performance story

- Compose-first UI with restrained animation budgets for mid-range MediaTek devices.
- Same APK path for Snapdragon flagships — no vendor-specific forks required for MVP.
- Adaptive layouts for tablets (width windows) without a separate tablet codebase.
- Baseline profile / R8 ready for release builds.

7. Open Source & Install

Distribution target: **GitHub Releases** APK. Users enable Unknown Sources / sideload, install SpideyOS, optionally set as default launcher, and grant Notification access + Display over other apps for the full Spidey experience.

- No root / Magisk required for core features.
- Works across OEM skins that ship Android 15+.
- MIT-style code license with clear Marvel non-affiliation disclaimer.
- Install guide and architecture docs live in the repository.

8. What I Built & Learned

SpideyOS is a portfolio-grade demonstration of end-to-end Android product engineering:

- Kotlin & modern Android (API 35)
- Jetpack Compose UI + custom motion (web motifs)
- System integration: NotificationListenerService, launcher HOME intent, overlays
- Generative AI product design (Gemini persona, latency-aware UX)
- CameraX custom camera experience
- Cross-SoC thinking (Snapdragon + MediaTek) and tablet adaptivity
- Open-source release engineering & technical storytelling

9. Call to Action

Star the repo. Install the APK. Tell me what Spidey should say next.

GitHub: [YOUR_GITHUB_URL](#)

Author: Jaikar

Open to Android / mobile / AI product engineering conversations.

Disclaimer

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With great power comes great responsibility — and great commit messages.